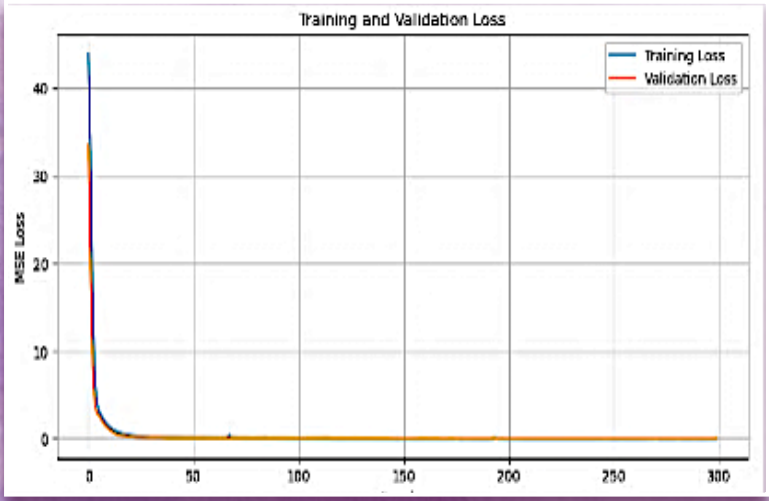
FLOOD DEFENCE

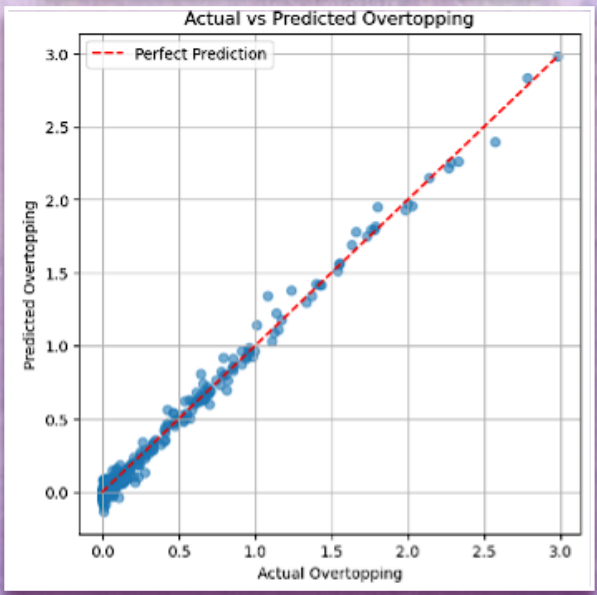
DATA

Design ID	R_c	Alpha	Year
3	1.08	77.4	2009
10	4.84	56.3	2010
7	1.37	10	2012
14	4.55	85.8	2015
K2017	3.5	50	2017
20	5.42	18.4	2018

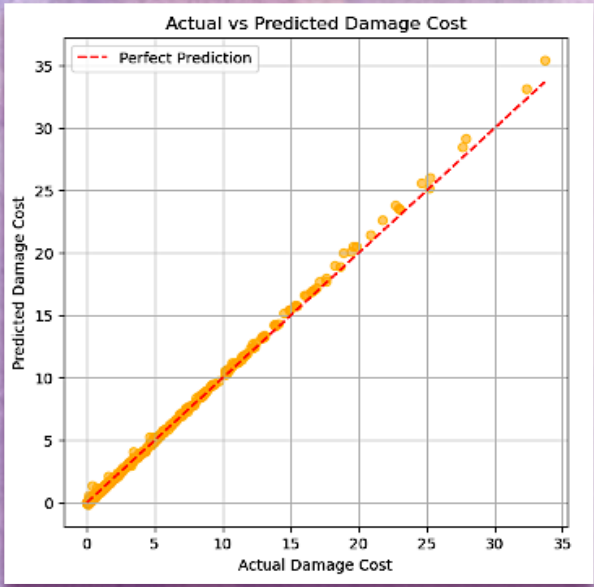
ACCURACY



DATA CLEANING: NaN



DATA TRAINING: ANN



BLACKBOX - We have coded our own neural netwrok and trained our model to make accurate predictions of damage and overtopping costs by using data from previous years (2009-2018) which makes it more trustworthy and reliable.

DATA RANGE - The design parameters our model involves are the R_c and alpha values which show the building and damage costs. The range of environmental conditions our model can handle ranges from 0-3 overtopping, ensuring affordability but also a flood defence which can withstand harsh conditions.

USABILITY - Our model is easy to use as it shows graphs for the overtopping and damage costs possesing computational efficiency and any change in data does not require a rewrite of the entire code.

OUR DESIGN TO SELL - To conclude, our usable model ensures a flood defence that can withstand overtopping conditions, based on predicitons made from previous years but also ensures affordability and a reasonable price too.